



Multiple Choice:

1. What is the process...

Ans. **C. Annealing**

2. Which of the following...

Ans. **C. Iron (Fe)**

3. Which of the following...

Ans. **D. Polyether ether ketone (PEEK)**

4. Which type of...

Ans. **C. Metallic Bonding**

5. What type of material is...

Ans. **C. Single Crystal Material**

6. What is the process by...

Ans. **D. Compression testing**

7. Which alloying element is ...

Ans. **A. Chromium (Cr)**

8. Which type of heat treatment...

Ans. **D. Quenching**

9. Which type of bonding is ...

Ans. **B. Ionic Bonding**

10. Which of the following...

Ans. **C. Hardness testing**

11. What is the critical cooling...

Ans. **D. Extremely rapid cooling**

12. Which of the following properties...

Ans. **D. Ductility**

13. Which type of microscopy...

Ans. **B. Scanning electron microscopy (SEM)**

14. Which type of material has a...

Ans. **A. Amorphous material**

15. Which type of bonding is ...

Ans. **A. Covalent Bonding**

16. Which material is commonly...

Ans. **D. Carbon fibers**

17. Which material exhibits shape...

Ans. **D. Nitinol**

18. Which material is commonly...

Ans. **D. Platinum**

19. Which material is commonly...

Ans. **C. Titanium**

20. Which material is commonly ...

Ans. **C. Graphite**

21. Which material is commonly...

Ans. **C. Gallium Arsenide**

22. Which material is commonly ...

Ans. **D. Yttrium Barium Copper Oxide (YBCO)**

23. Which material is commonly...

Ans. **C. Zinc**

24. Which of the following ...

Ans. **C. Cellulose**

25. Which polymer is commonly...

Ans. **A. Polyethylene**

26. Which of the following polymers...

Ans. **C. Polyimide**

27. Which polymer is commonly...

Ans. **C. Polylactic acid (PLA)**

28. Which of the following ...

Ans. **D. Polylactic Acid (PLA)**

29. Which nanomaterial exhibits...

Ans. **C. Silicon nanoparticles**

30. Which nanomaterial is...

Ans. **B. Iron oxide nanoparticles**

31. Martensite is a microstructure...

Ans. **B. Quenching of a material**

32. The formation of...

Ans. **D. Martensitic transformation**

33. Martensite formation...

Ans. **B. Hardness**

34. The presence of alloying ...

Ans. **A. Austenite**

35. Which of the following phases...

Ans. **A. Austenite**

36. Pearlite is a lamellar ...

Ans. **B. Ferrite and Cementite**

37. Austenite is the stable phase ...

Ans. **B. Above the eutectoid temperature**

38. The transformation of austenite...

Ans. **B. Normalizing**

39. Which phase is formed ...

Ans. **D. Martensite**

40. Which of the following...

Ans. **B. Precipitation hardening**

41. Which strengthening...

Ans. **A. Work hardening**



42. Which strengthening process...

Ans. **C. Solid solution strengthening**

43. Which strengthening process...

Ans. **D. Grain refinement**

44. Which strengthening process...

Ans. **D. Point defect strengthening**

45. The arrangement of atoms...

Ans. **C. Size**

46. The crystal lattice structure...

Ans. **A. Cubic**

47. The crystal lattice structure...

Ans. **C. Hexagonal**

48. Which crystal lattice structure...

Ans. **C. Face-Centered Cubic (FCC)**

49. The crystal lattice structure...

Ans. **A. Cubic**

50. Which of the following...

Ans. **B. X-Ray Diffraction**